

Unicorn Business

Global Unicorn Company Analysis — Full Project Documentation

Python · MySQL · Power BI Desktop

1,074 Companies · \$3.7T Combined Valuation · 46 Countries

TOOLS	SOURCE	RECORDS	SCOPE
Python · MySQL · Power BI	Global Unicorn Companies Dataset	1,074 · 13 Cols	46 Countries, 6 Continents

01 EXECUTIVE SUMMARY

- **Analyzed 1,074 global unicorn companies** (private startups valued at \$1B+) across 46 countries — a dataset clean of duplicates but requiring real parsing work: Valuation and Funding stored as formatted text ('\$180B', '\$572M'), a capitalization split fracturing one industry category into two, and a comma-separated investor list hiding a genuine many-to-many relationship inside a single text column.
- **Answered 7 structured questions** covering valuation leaders, industry concentration, country and continent distribution, the funding-valuation relationship, and time-to-unicorn patterns — then reproduced every answer independently in MySQL across a properly normalized two-table schema.
- **Built a 3-page, 12-visual interactive Power BI dashboard** with 4 DAX measures and a dedicated Investors page powered by a real junction table — not a workaround, but the correct relational structure for a many-to-many relationship.
- **Surfaced a sharp geographic concentration story:** the United States alone accounts for 562 of 1,074 unicorns (52.3%) and North America holds \$2.03T of the ~\$3.7T combined valuation — more than double Asia's \$1.07T and vastly ahead of Africa's \$5B, a 400x gap between the top and bottom continent.

02 PROBLEM STATEMENT

The unicorn companies dataset was distributed as a flat, human-readable export — formatted for reading, not for analysis. Valuation and Funding were stored as display strings with currency symbols and magnitude suffixes rather than numbers; one genuine data category (Artificial Intelligence) was silently split into two by an inconsistent capitalization convention; and the Select Investors field packed an unbounded list of investor names into a single comma-separated string, which is structurally a many-to-many relationship (one company can have several investors, one investor backs many companies) masquerading as a flat text field. Two companies also happened to share the exact name 'Bolt' despite being entirely unrelated businesses, a coincidence that risked being mishandled as a duplicate. The goal of this project was to correctly parse, deduplicate the category system, and properly normalize the investor relationship before answering 7 concrete questions about where unicorn value is concentrated.

03 OBJECTIVES

- **Obj 1:** Parse Valuation and Funding from formatted display strings into consistent numeric values (\$ millions).
- **Obj 2:** Detect and merge the Industry capitalization duplicate before any industry-level ranking.
- **Obj 3:** Correctly distinguish a genuine same-name coincidence (two different 'Bolt' companies) from an actual duplicate record.
- **Obj 4:** Normalize the Select Investors field into a proper many-to-many junction table rather than leaving it as an unparseable text blob.
- **Obj 5:** Reproduce all 7 answers independently in MySQL across a two-table schema with a real foreign-key relationship.
- **Obj 6:** Build a Power BI dashboard with a dedicated page that puts the investor junction table to genuine analytical use.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Global Unicorn Companies Dataset
File Format	CSV, single flat export
Records	1,074 companies → normalized into companies (13 cols) + company_investors (2 cols, 3,051 rows)
Geographic Scope	46 countries · 6 continents
Duplicate Rows	Zero — including 'Bolt', confirmed as two distinct companies via differing investor lists
Missing Values	City: 16 nulls (all city-state countries) · Select Investors: 1 null · Funding: 12 'Unknown' strings
Category Defect	Industry: 'Artificial Intelligence' (11 rows) vs. 'Artificial intelligence' (73 rows) — same category
Data-Integrity Error	1 row (Yidian Zixun): Year Founded (2021) is after Date Joined unicorn status (2017) — flagged, not corrected

Key Columns Used in Analysis

Column	Type	Description
company	Text	1,074 unique company names (2 rows share 'Bolt', genuinely different companies)
valuation_m	Decimal (derived)	Parsed from text (\$180B → 180,000); company valuation in \$ millions
funding_m	Decimal (derived)	Parsed from text; 'Unknown' → true null, not zero
industry	Text	15 categories after merging the capitalization duplicate
country / continent	Text	46 countries across 6 continents
years_to_unicorn	Integer (derived)	Date Joined year – Year Founded; nulled for the 1 impossible row
investor	Text (junction table)	1,253 unique investors, normalized from the comma-separated source column

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Python 3	VS Code	Text-to-numeric parsing, category merging, list normalization
pandas / numpy	Python library	String parsing with unit-suffix logic, junction-table construction via row explosion
MySQL 8	MySQL Workbench	Two-table relational schema with a real one-to-many foreign key; SQL validation
Power BI Desktop	Microsoft	3-page dashboard including a dedicated Investors page built on the junction table

Tool	Platform	Purpose in This Project
DAX	Power BI native language	4 measures: Total Companies, Total Valuation, Avg Years to Unicorn, Total Investors

06 DATA CLEANING PROCESS — PYTHON

No duplicates, but four distinct parsing and normalization problems requiring deliberate handling.

Step	Action	Result
1	Parse Valuation/Funding text ('\$180B' → 180000, '\$572M' → 572); 'Unknown' → NaN first	Two clean numeric columns in \$ millions
2	Merge 'Artificial Intelligence' into 'Artificial intelligence'	One clean 84-company category instead of two fragments (11+73)
3	Parse Date Joined to true date; compute Years_To_Unicorn	Derived tenure metric, nulled for 1 logically-impossible row
4	Confirm the 'Bolt' pair are genuinely different companies (differing investors/countries), not deduped	2 legitimate same-name records correctly preserved
5	Explode Select Investors on commas into a company_investors junction table	3,051 company-investor relationship rows, 1,253 unique investors

07 SQL / MYSQL — VALIDATION LAYER

A properly normalized two-table schema: companies (1,074 rows, one per unicorn) and company_investors (3,051 rows, one per company-investor relationship), linked by a real foreign key on company_id — the correct relational structure for what was originally a many-to-many relationship jammed into a single text column.

Two Nullable Columns, Handled Proactively

funding_m and years_to_unicorn both have genuine nulls (12 'Unknown' fundings, 1 flagged bad-data row). Both use explicit NULLIF(@raw,") in the LOAD DATA statement from the first import, applying the lesson from an earlier project in this series where this exact pattern silently loaded empty fields as 0 instead of NULL.

MySQL's Missing Correlation Function

The funding-valuation relationship (Q5) was implemented with the same hand-built Pearson formula used throughout this project series, since MySQL has no native CORR(). Result: 0.5984, matching Python exactly.

08 DAX MEASURES CREATED

Measure	DAX Formula	Result
Total Companies	COUNTROWS(companies)	1,074
Total Valuation	SUM(companies[valuation_m])	~\$3.7 trillion (combined)
Avg Years to Unicorn	AVERAGE(companies[years_to_unicorn])	7.0 years (median 6)
Total Investors	DISTINCTCOUNT(company_investors[investor])	1,253

09 DASHBOARD VISUALS BUILT (12 Visuals, 3 Pages)

Page 1 — Global Overview

#	Visual Type	Fields Used	Insight Delivered
1-3	KPI Cards	Total Companies / Total Valuation / Avg Years to Unicorn	1,074 companies, ~\$3.7T combined, 7.0-year average tenure
4	Bar Chart	Y: company / X: valuation_m, Top 10	Bytedance leads at \$180B
5	Filled Map	Location: country / Color: Total Valuation	US and China dominate visually
6-7	Slicers	continent, industry	Cross-filters all Page 1 visuals

Page 2 — Industry & Funding

#	Visual Type	Fields Used	Insight Delivered
1	Clustered Bar (x2)	industry by Total Companies, and separately by Total Valuation	Fintech leads both rankings (224 companies, \$882B)
2	Scatter + trend line	X: funding_m / Y: valuation_m (Don't summarize)	Moderate positive relationship, corr = 0.598
3	Histogram (binned column)	years_to_unicorn distribution	Right-skewed; median 6, long tail out to legacy companies
4	Slicer	industry	Filters both bar charts and the scatter

Page 3 — Investors

- Bar Chart (from company_investors): Top 10 investors by number of companies backed — Accel leads at 60, followed by Tiger Global Management and Andreessen Horowitz (53 each).
- Table: investor-driven drill-down showing which specific companies a slicer-selected investor has backed, joined back to companies for valuation context.
- Slicer: investor (searchable dropdown, 1,253 values) — this entire page only exists because the data was normalized into a real junction table rather than left as an unparseable list.

10 KEY INSIGHTS & RESULTS

A. Valuation Leaders

- **Bytedance leads all unicorns at \$180B**, nearly double the next tier (SpaceX and SHEIN, both \$100B) — a genuinely outsized #1 rather than a close race at the top.

B. Industry Concentration

- **Fintech dominates on both dimensions:** 224 companies (the largest count) and \$882B combined valuation (the largest total) — the count and value rankings agree at the top, though they diverge further down the list.

C. Geographic Concentration

- **The United States alone holds 52.3% of all global unicorns** (562 of 1,074), more than triple China's count (173) and nearly nine times India's (65).

- **North America's valuation lead is even starker than its company-count lead:** \$2.03T versus Asia's \$1.07T (roughly 1.9x) and Europe's \$503B — while Africa holds just \$5B, a ~400x gap from the top continent.

D. Funding & Time-to-Unicorn

- **Funding and valuation are meaningfully but not tightly linked:** $r = 0.598$ — more capital raised associates with higher valuation, but plenty of high-valuation companies raised comparatively little, and vice versa.
- **The typical company reaches unicorn status in about 6 years**, though the distribution has a long tail: two companies founded before 1980 (Otto Bock HealthCare in 1919, Promasidor Holdings in 1979) only reached unicorn status decades later — a fundamentally different story from the typical fast-growth tech narrative.

E. Investor Landscape

- **Accel is the most prolific unicorn investor by count**, backing 60 companies in this dataset, narrowly ahead of Tiger Global Management and Andreessen Horowitz (53 each) — visible only because the investor list was properly normalized into its own table.

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Numbers Formatted for Reading, Not Analysis

Problem: Valuation and Funding were stored as display strings ('\$180B', '\$572M') with inconsistent magnitude suffixes, unusable directly in any aggregation or correlation.

Solution: Built a parsing function that strips the currency symbol, detects the B/M suffix, and converts to a single consistent unit (\$ millions) — applied after first converting the literal string 'Unknown' to a true null, so it wouldn't be misparsed as a numeric zero.

Challenge 2 — One Category Split in Two by Capitalization

Problem: 'Artificial Intelligence' and 'Artificial intelligence' were treated as separate categories by any naive groupby, splitting a true 84-company category into fragments of 11 and 73.

Solution: Checked `.unique()` on every categorical column during assessment specifically looking for near-duplicate strings, not just exact duplicates — caught and merged before any industry-level ranking.

Challenge 3 — A Same-Name Coincidence That Looked Like a Duplicate

Problem: 'Bolt' appears twice in the company list, which a naive `.duplicated()` check on Company alone would flag for removal.

Solution: Inspected the full rows behind the apparent duplicate before acting — different countries, different investor lists confirmed these are two unrelated real companies that happen to share a name, not a data error. Preserved both.

Challenge 4 — A Many-to-Many Relationship Trapped in One Text Column

Problem: Select Investors packed an arbitrary-length, comma-separated investor list into a single field — unusable for any investor-level question ('which investor backs the most unicorns?') without restructuring.

Solution: Exploded the column into a proper `company_investors` junction table during cleaning, one row per company-investor pair — turning an unanswerable question into a straightforward GROUP BY, and giving the Power BI dashboard a genuine third page it wouldn't otherwise have had.

12 SKILLS LEARNED & TAKEAWAYS

- Parsing formatted display strings into analysis-ready numerics, including handling a literal 'Unknown' placeholder correctly before any unit conversion logic runs.
- Auditing categorical columns for near-duplicate variants (capitalization, spacing) specifically, not just exact-match duplicates.
- Verifying a suspected duplicate against the full row context before removing it — a same-name coincidence and a true duplicate look identical on a single-column check.
- Recognizing when a delimited text field is actually a disguised many-to-many relationship, and normalizing it into a proper junction table rather than working around it with string-splitting logic at query time.
- Applying the NULLIF lesson from an earlier project proactively at first import, rather than discovering the same empty-field-becomes-zero bug a second time.
- Building a dashboard page that only exists because of a specific normalization decision — the Investors page is a direct payoff of the junction-table design, not an afterthought.

13 CONCLUSION

- **Parsed and normalized 1,074 unicorn company records**, converting display-formatted valuation and funding text into analysis-ready numerics, merging a fractured industry category, and correctly distinguishing a same-name coincidence from a true duplicate.
- **Independently validated all 7 project questions in MySQL** across a properly normalized two-table schema with a real many-to-many relationship resolved into a junction table — not a workaround, the correct relational design.
- **Delivered a 3-page, 12-visual Power BI dashboard**, including a dedicated Investors page that exists specifically because the data was normalized correctly rather than left as an unparseable list.
- **Most notably, quantified a stark concentration story**: the US holds over half of all global unicorns and North America's valuation share is nearly double Asia's and roughly 400x Africa's — a geographic imbalance visible only once the underlying data was made properly analyzable.

KPI Summary

Metric	Value	Metric	Value
Total Companies	1,074	Countries / Continents	46 / 6
Combined Valuation	~\$3.7 Trillion	Top Company	Bytedance — \$180B
Top Industry (Count)	Fintech — 224 companies	Top Industry (Value)	Fintech — \$882B
Top Country	United States — 562 (52.3%)	Top Continent (Value)	North America — \$2.03T
Funding ↔ Valuation Corr.	0.598	Avg Years to Unicorn	7.0 (median 6)
Total Investors	1,253	Top Investor	Accel — 60 companies
Investor Relationships	3,051	Industry Categories (merged)	15
Duplicates	0	City Nulls (structural)	16

